SSAVI-GMM: An Automatic Algorithm for Mapping Submerged Aquatic Vegetation in Shallow Lakes Using Sentinel-1 SAR and Sentinel-2 MSI Data

Yihao Xin, Juhua Luo, Ying Xu, Zhe Sun, Tianci Qi, Ming Shen, Yinguo Qiu, Qitao Xiao, Linsheng Huang, Jinling Zhao, and Hongtao Duan

***Abstract*— Submerged aquatic vegetation (SAV) is crucial for maintaining a clear-water state in lakes. Tracking the spatiotemporal changes in SAV is crucial for understanding the ecological evolution, particularly in eutrophic lakes. Sentinel imagery offers high-resolution data for detailed SAV mapping. However, existing SAV classification algorithms based on Sentinel-2 multispectral instrument (MSI) require preprocessing to eliminate interference from other types such as floating-leaved aquatic vegetation and algal blooms (ABs), and also heavily rely on field survey data and human interventions, limiting the application for large-scale and long-term SAV monitoring. Here, we developed SSAVI-GMM, an algorithm leveraging a novel index, the Sentinel-based SAV index (SSAVI), derived from Sentinel-1 synthetic aperture radar (SAR) and Sentinel-2 MSI data, to automatically map SAV distribution using Gaussian mixture model (GMM) clustering. Testing in 19 lakes in the Yangtze Plain region yielded an average accuracy of 87.15%. This study marks a successful integration of SAR and optical data, addressing challenges in mapping SAV, and the robust GMM clustering method overcomes the limitations of traditional threshold methods. The SSAVI-GMM algorithm demonstrates promising potential for mapping SAV in shallow lakes globally.**

***Index Terms*— Gaussian mixture model (GMM) clustering, image fusion, remote sensing, Sentinel data, submerged aquatic vegetation (SAV), synthetic aperture radar (SAR)–optical.**

I. Introduction

SUBMERGED aquatic vegetation (SAV), a primary producer in lake ecosystems, performs crucial ecological and environmental functions in lakes, including regulating nutrient levels, stabilizing sediments and enhancing water clarity, and providing food and habitats for organisms [1], [2], [3]. It also exerts allelopathic effects to inhibit algal growth, thus constituting a key mechanism underlying the induction and maintenance of clear-water state in lakes [4], [5]. However, recent studies revealed that global lacustrine SAV has experienced significant reductions based on literature data over the past few decades, attributed to climate warming and anthropogenic actions [6], [7], [8]. Whether this suggests that lakes are shifting from a clear-water state dominated by SAV to a turbid water state dominated by algal blooms (ABs) is becoming a hot topic in the field of ecology [9], [10]. In this context, the distribution, quantity, and high spatiotemporal variation information of SAV are critical for detecting lake state shifts, quantifying state transition thresholds, and informing future SAV restoration efforts.

Traditional field surveys for acquiring SAV distribution information often demand extensive time and labor resources. In addition, the limited survey dataset cannot adequately represent the intricate temporal and spatial heterogeneity of SAV distribution. In contrast, satellite remote sensing has offered advantages in terms of wide coverage, regular revisits, and long-term observation, making it an effective approach to address the issue of missing regional or global-scale SAV distribution data in lakes. Based on optical remote sensing, an array of spectral indices were developed for mapping SAV, such as submerged vegetation spectral index (SVSI) [11], cyanobacteria and macrophytes index (CMI) [12], normalized difference aquatic vegetation index (NDAVI), water adjusted vegetation index (WAVI) [13], and floating algae index (FAI) [14]. In lakes without floating-leaved and emergent

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aquatic vegetation (FEAV) and AB, theoretically, based on these indices, SAV can be directly extracted. However, in lakes with AB and FEAV, the traditional methods for SAV extraction involve constructing a decision tree to remove FEAV and AB using additional spectral features such as NDVI before extracting SAV by these indices. This is because the spectra of SAV are influenced by both the water column and the canopy, exhibiting characteristics similar to those of FEAV and AB in certain bands. The spectral indices constructed based on these bands may also be sensitive to FEAV and AB [14], [15], [16]. This approach faces two significant challenges for mapping SAV in such lakes, particularly in eutrophic lakes where SAV, FEAV, and AB coexist. First, this multilayer tree structure, formed by the need to eliminate various interfering information, is susceptible to the accumulation and propagation of errors, leading to increased uncertainty in mapping SAV [17], [18]. In addition, thresholds of those indices in the decision trees highly rely on field survey data or manual interventions, which limit the application of large-scale and long-term monitoring [19]. Although some studies developed algorithms with dynamic image-specific thresholds based on linear mixing models in certain studies [13], [14], [15], they still require field survey samples for determining the mixing ratios of lake cover types. Overall, optical remote sensing algorithms for mapping SAV face challenges in terms of both multiple spectral indices and threshold determination. In contrast to optical remote sensing, synthetic aperture radar (SAR) can capture volume scattering and surface scattering through different polarization directions. The polarization bands of SAR show sensitivity to vegetation inundation conditions and canopy structures, making it widely applied in mapping FEAV in lakes or wetlands [20], [21]. However, there is currently limited research on the mechanisms and applications of SAR for SAV mapping. SAR can provide information on surface roughness, dielectric constant, humidity, and morphology of objects [22], which are challenging to obtain through optical remote sensing. This additional information provided by SAR serves as a valuable supplement to optical data. And numerous studies have demonstrated that the integration of SAR and optical data can significantly improve the classification accuracy of various land cover types, particularly wetland vegetation communities [23], [24], [25]. This fusion approach has also been applied to water quality parameter studies, proving the responsiveness of SAR to the properties of the water surface [26], [27]. Therefore, can we address above challenges of mapping SAV by fusing the information from SAR and canopy information from optical remote sensing?

The Sentinel series of satellites in the Copernicus program initiated by the European Space Agency (ESA) provide Sentinel-1 SAR and Sentinel-2 multispectral instrument (MSI) data, which have excellent data complementarity of land cover mapping [28], [29]. Furthermore, the spatial resolutions of Sentinel-1 SAR ground range detected (GRD) and Sentinel-2 MSI data are 10 m, respectively, with maximum revisit intervals of six and five days, rendering them optimal for observations of SAV high spatiotemporal changes.

Therefore, we aim to develop an algorithm for mapping SAV with low dependency and high universality by fusing Sentinel-1 SAR and Sentinel-2 MSI data. Specific objectives are: 1) to construct a new index that is only sensitive to SAV but not to FEAV and AB by fusing Sentinel-1 SAR and Sentinel-2 MSI and 2) to develop an automatic classification algorithm for mapping SAV with threshold independent.

II. Materials

A. Study Area

We selected 19 lakes in the Yangtze Plain to develop and validate SAV mapping algorithm (Fig. 1 and Table S1). Among them, Lake Taihu is a typical eutrophic lake with coexistence of SAV, FEAV, and AB [Fig. 1(e)]. The western part of Lake Taihu frequently experiences ABs, characterizing it as a typical algae-dominated zone, while the eastern part is home to extensive SAV, making it a typical macrophyte-dominated zone [30]. Therefore, Lake Taihu is a representative experimental lake to develop the algorithm for mapping SAV. In addition, we conducted detailed validation in five lakes, including Lake Taihu, Lake Datong, Lake Shijiu, Lake Nanyi, and Lake Yangcheng [Fig. 1(b)–(f)]. And the other 14 lakes were used for extensive validation of SAV mapping algorithm.

B. Field Data

The field dataset consists of three parts, and we provided detailed information about lakes and three datasets (Tables S1 and S2). Dataset A was collected by DJI Phantom 4 RTK and boat in Lake Taihu for developing algorithm from August 21 to 29 in 2019. Specifically, a total of 19 sample regions were collected, including eight regions for FEAV, two regions for AB, three regions for OW, and six regions for SAV (Fig. S1). Two of SAV sample regions were acquired by DJI Phantom 4 RTK, and all the others were collected by boat marking geotagged polygons where target communities were present. Simultaneously, Sentinel-2 imagery was acquired on August 17, 2019. Using these sampled locations, regions of interest were discerned on Sentinel-2 imagery. Subsequently, a total of 2104 SAV pixels, 2181 OW pixels, 1623 FEAV pixels, and 1752 AB pixels were individually extracted from this imagery to constitute dataset A.

We collected 384 samples in five lakes through field survey as dataset B for detailed validation. Specifically, during the period from March 10 to 15, 2018, 45 SAV samples were acquired in Lake Shijiu. From August 1 to 7, 2018, 56 SAV samples were yielded in Lake Nanyi. In all, 49 SAV samples were acquired in Lake Datong from June 5 to 12, 2019. In Lake Yangcheng, 54 SAV samples were yielded from March 10 to 15, 2022. We also acquired 61 SAV samples in Lake Taihu from September 3 to 6, 2020, and 119 SAV samples from July 10 to 18, 2022.

Dataset C constructed by 757 SAV samples was collected by literature retrieval and visual interpretation from 16 lakes.

C. Satellite Images

We used Sentinel-1 and Sentinel-2 data from the Sentinel series of the ESA in this study. Of these, Sentinel-1 carries C-band SAR payload operating at 5.405 GHz and the

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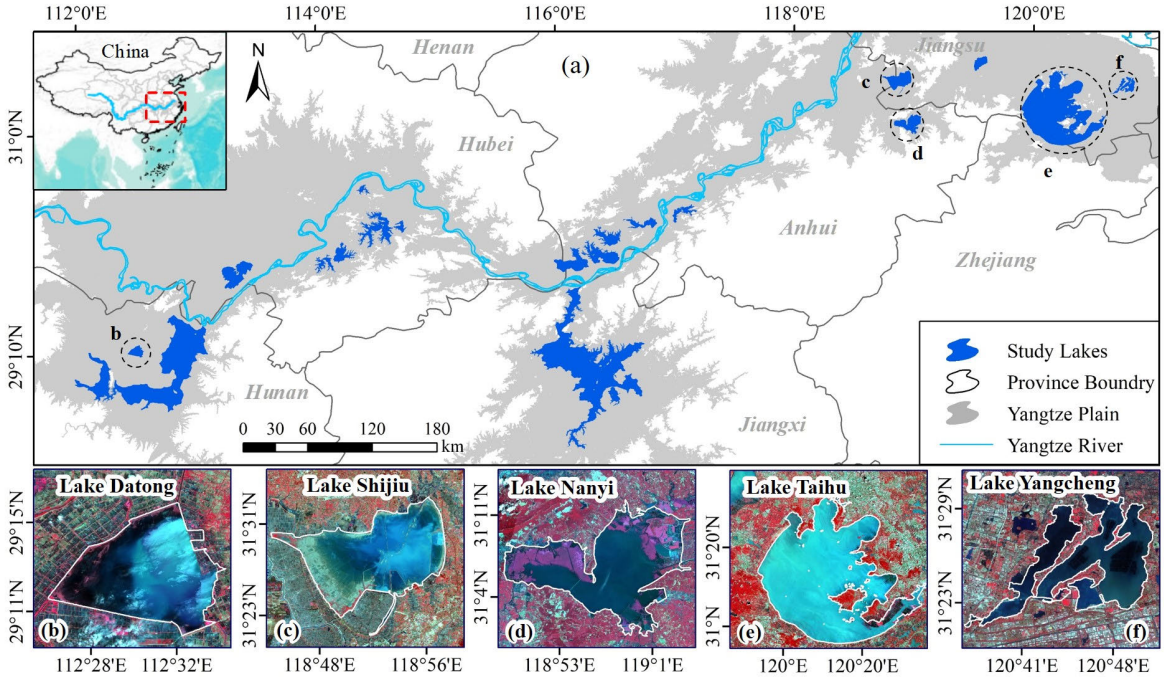


Fig. 1. Location of study lakes. (a) Distribution of 19 study lakes and (b)–(f) five detailed validation lakes in the Yangtze Plain.

maximum revisit interval is six days. In this study, we used interferometric wide swath (IW) mode of Sentinel-1, transforming data through processing steps including ground range detection, orbit correction, terrain correction, and thermal noise reduction, resulting in 10-m ground-projected images. The openly accessible dataset interface for Sentinel-1 SAR GRD data in Google Earth Engine (GEE) is referenced as "COPERNICUS/S1_GRD."

Sentinel-2 main payload is an MSI with 13 spectral bands and the minimum revisit interval is five days. GEE offers two data products: Level-1C Top of Atmosphere (TOA) and Level-2A Bottom of Atmosphere (BOA). The Sentinel-2 Level-1C TOA dataset undergoes preliminary processing involving radiometric calibration and geocorrection and is often used for SAR data fusion [31], [32]. In this study, the Sentinel-2 Level-1C TOA dataset was used and the openly accessible dataset interface for this information is denoted as "COPERNICUS/S2."

In addition, we also acquired submeter high-resolution JL-1 satellite images on October 4, 2021 in Lake Taihu, September 22, 2021 in Lake Nanyi, June 7, 2021 in Lake Yangcheng, and October 22, 2020 in Lake Wuchang (www.jl1mall.com). These images consist of five bands [pan, blue, green, red, near-infrared (NIR)] with a spatial resolution of 0.7 m. SAV is clearly visible in the true-color images, and the SAV distribution can be obtained through visual interpretation to assess our algorithm.

III. Method

We developed an automatic algorithm for mapping SAV, named SSAVI-GMM. It included two main steps: 1) construction of a new index, named Sentinel-based SAV index (SSAVI), by fusion of the Sentinel-1 SAR and Sentinel-2 MSI imagery on GEE and 2) extraction of SAV by combining SSAVI and Gaussian mixture model (GMM) clustering in the local Python environment (Fig. 2).

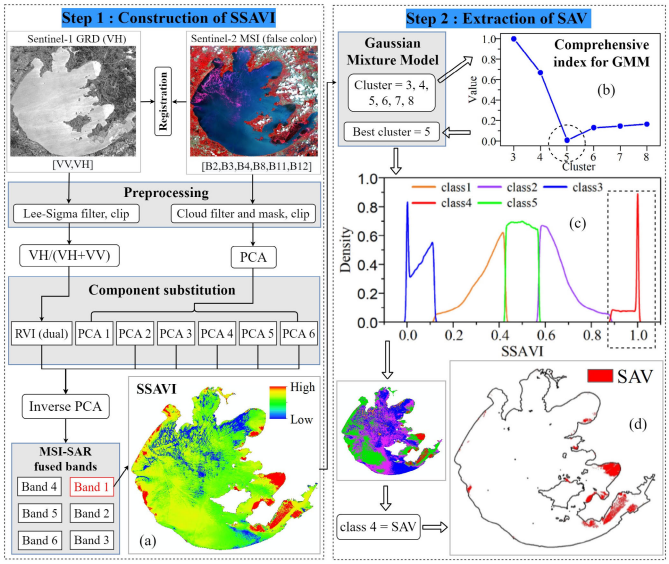


Fig. 2. Workflow of SSAVI-GMM algorithm proposed for mapping SAV based on a case in Lake Taihu. In this case, Sentinel-2 MSI TOA and Sentinel-1 SAR GRD images were acquired on July 28, 2018 and August 5, 2018, respectively. (a) SSAVI, (b) comprehensive index of different cluster numbers, (c) distribution of SSAVI value of each class in the optimal clustering result, and (d) map of SAV.

A. Construction of SSAVI

The SSAVI was constructed by fusing the Sentinel-1 SAR and Sentinel-2 MSI imagery on GEE. Specifically, Sentinel-2 MSI TOA and Sentinel-1 SAR GRD imagery with the closest time interval to Sentinel-2 were acquired and preprocessed. Preprocessing of the Sentinel-2 MSI TOA involved cloud masking of Sentinel-2 data by quality assessment band named QA60 and image filtering of Sentinel-1 data

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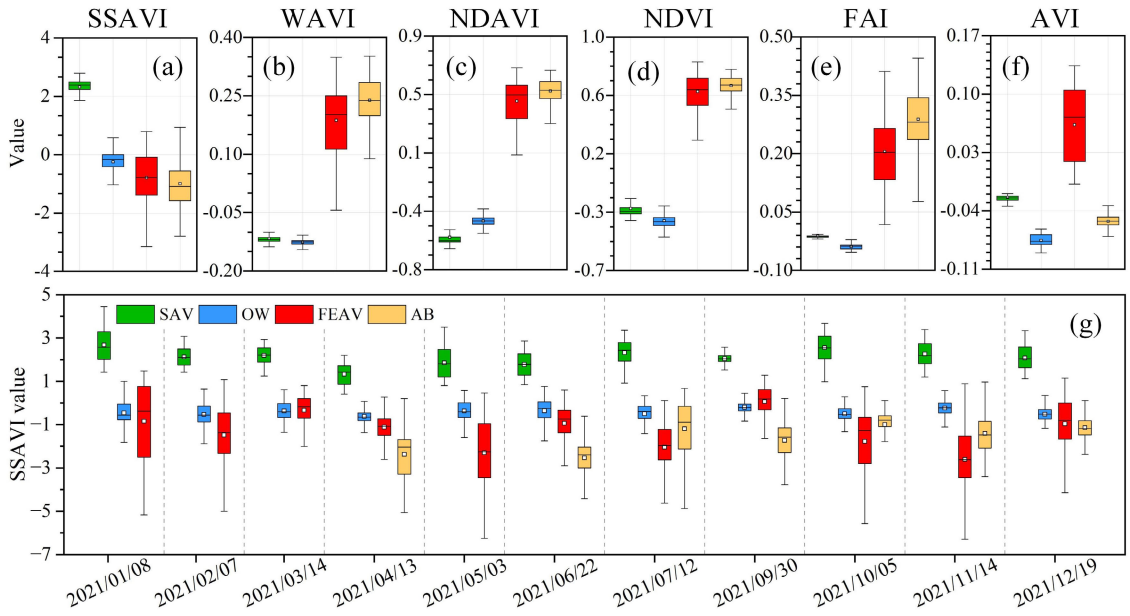


Fig. 3. Sensitivity and robustness of SSAVI for SAV, taking Lake Taihu as an example. Value distribution of SAV and other types (FEAV, AB, and OW) in (a)–(f) SSAVI, WAVI, NDAVI, NDVI, FAI, and AVI. (g) Value distribution of SAV and other types in SSAVI at each month of 2021.

using improved Lee Sigma filter [33]. A dual-polarization radar vegetation index ($\mathrm{RVI_{dual}}$) was computed based on vertical–vertical (VV) and vertical–horizontal (VH) polarization bands of Sentinel-1 (1). The $\mathrm{RVI_{dual}}$ possesses a similar structure to RVI [34]. Then, the principal component analysis (PCAs) was conducted for the Sentinel-2 MSI image with band blue, green, red, NIR, short-wave infrared 1 (SWIR 1), and short-wave infrared 2 (SWIR 2) (i.e., B2, B3, B4, B8, B11, and B12), and a transformation coefficient matrix X was obtained by feature vector function of GEE (Decomposition-Factory.eig() from https://ejml.org.). Subsequently, based on a pixel-level image fusion approach called principal component substitution (PCS) [35], [36], we replaced the first principal component, which has the highest information content, with $\mathrm{RVI_{dual}}$. And as the inverse transformation coefficient matrix, the transpose matrix X^{T} of X was calculated. Finally, using the replaced components and X^{T}, we obtained six fused bands that incorporate both SAR and optical information (2) and (3). Among these, we observed that SAV is concentrated in high-value areas, while OW, FEAV, and AB are distributed in low-value areas on the image of the first band after fusing (Band 1 in Fig. 2), so we designated Band 1 as SSAVI [Fig. 2(a)]

$$\mathrm{RVI_{dual}} = \frac{\sigma_{\mathrm{VH}}}{\sigma_{\mathrm{VV}} + \sigma_{\mathrm{VH}}} \tag{1}$$

$$\mathrm{SSAVI} = \mathrm{RVI} \times X_{11} + [R_{\mathrm{Blue}}, R_{\mathrm{Green}}, R_{\mathrm{Red}}, R_{\mathrm{NIR}}, R_{\mathrm{SWIR1}}, R_{\mathrm{SWIR2}}] \times Y \tag{2}$$

$$Y = \left[X_{ij}\right] \times \left[X_{1j}\right]^{\mathrm{T}} \tag{3}$$

where σ_{VH} and σ_{VV} are the backscattering coefficients for VH and VV polarizations, respectively. Y is the matrix of weight coefficients that make up the components in SSAVI. R is the reflectance and the subscript is the spectral band. X is the 6 × 6 matrix of rotation coefficients produced by PCA transformation, with subscripts representing rows and columns, respectively, $i = (1, 2, 3, 4, 5, 6)$ and $j = (2, 3, 4, 5, 6)$.

We further tested the sensitivity and robustness of SSAVI in detecting SAV. First, the separability of SSAVI for SAV was assessed by comparing with the commonly used optical spectral indices for mapping aquatic vegetation, including NDVI [37], FAI [14], [38], AVI [15], WAVI, and NDAVI [39], [40] (4)–(8) [Fig. 3(a)–(f)]. Among these, NDVI, FAI, WAVI, and NDAVI all displayed excellent separation capabilities for the cover types that have typical spectral characteristics of vegetation, such as FEAV and AB [Fig. 3(b)–(e)], but failed to extract SAV from OW [Fig. 3(b)–(e)]. Although AVI could differentiate both OW and SAV, as well as FEAV and AB, additional indices such as NDVI were required for extraction of SAV [Fig. 3(f)]. In contrast, SSAVI showed its advantage of high sensitivity to SAV, with the highest value and enabling direct separate SAV from other cover types [Fig. 3(a)]. In addition, the robustness of SSAVI in detecting SAV was also validated based on 11 time-series images from Lake Taihu in 2021. The results indicated that the SSAVI values for SAV consistently remained the highest value and exhibited high separability from other cover types

$$\mathrm{NDVI} = \frac{R_{\mathrm{NIR}} - R_{\mathrm{Red}}}{R_{\mathrm{NIR}} + R_{\mathrm{Red}}} \tag{4}$$

$$\mathrm{FAI} = R_{\mathrm{NIR}} - R_{\mathrm{Red}} - (R_{\mathrm{SWIR1}} - R_{\mathrm{Red}}) \times \frac{\lambda_{\mathrm{NIR}} - \lambda_{\mathrm{Red}}}{\lambda_{\mathrm{SWIR1}} - \lambda_{\mathrm{Red}}} \tag{5}$$

$$\mathrm{AVI} = [R_{\mathrm{Blue}}, R_{\mathrm{Green}}, R_{\mathrm{Red}}, R_{\mathrm{NIR}}, R_{\mathrm{SWIR1}}, R_{\mathrm{SWIR2}}] \times (-k) \tag{6}$$

$$\mathrm{WAVI} = 1.5 \times \frac{R_{\mathrm{NIR}} - R_{\mathrm{Blue}}}{R_{\mathrm{NIR}} + R_{\mathrm{Blue}} + 0.5} \tag{7}$$

$$\mathrm{NDAVI} = \frac{R_{\mathrm{NIR}} - R_{\mathrm{Blue}}}{R_{\mathrm{NIR}} + R_{\mathrm{Blue}}} \tag{8}$$

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where R is the reflectance, $\boldsymbol{\lambda}$ is the wavelength of the band, and k is the wetness coefficient of tasseled cap in different satellites.

It is noteworthy that we used the Sentinel-2 MSI TOA rather than the Sentinel-2 MSI BOA data when constructing SSAVI, because the SSAVI based on TOA performed better in identifying SAV than BOA data by our tests (Fig. S2). In addition, we also attempted to fuse some common SAR features with optical data for constructing SSAVI. As a result, we found that the SSAVI based on $\mathrm{RVI}_{\mathrm{dual}}$ outperformed other features in identifying SAV (Fig. S3).

B. Extraction of SAV

Based on SSAVI, we used GMM clustering provided by the sklearn library of Python to further extract SAV. GMM clustering is a probabilistic model-based and unsupervised method, and it is widely used for classification of remote sensing images [41]. In this study, we used a robust expectation and maximization (EM) clustering algorithm of GMM for automatically extracting SAV [42]. Specifically, we assumed a range of clustering parameters from three to eight based on its general convergence range and conducted clustering sequentially. Each specific step of clustering involved: 1) initializing the parameters of Gaussian distributions for each cluster; 2) calculating the probability of each sample belonging to each Gaussian distribution and updating the parameters; 3) computing the logarithm of the model likelihood, stopping when convergence or the predetermined iteration limit is reached, otherwise repeating step 2); and 4) assigning each sample to the Gaussian distribution with the highest probability, thereby completing clustering.

Davies–Bouldin index (DBI) and Bayesian information criterion (BIC) were used to evaluate the clustering results and determine the optimal clustering parameter. The DBI measured the intracluster similarity and intercluster differences, with a smaller DBI indicating better clustering performance [43]. And the BIC was often used in model selection, with lower values indicating lower model complexity and better predictive performance [44]. In this study, these two metrics were normalized and combined to form a comprehensive index (CI) (9)–(11), where the lowest CI value corresponds to the optimal number of clusters [Fig. 2(b)]

$$\mathrm{DBI} = \frac{1}{k}\sum_{i=1}^{k}\max_{i\neq j}\left(\frac{\sigma_i+\sigma_j}{d\left(c_i,c_j\right)}\right) \tag{9}$$

where k is the number of clusters, C_x is the center of cluster x, σ_i is the average distance between the data in cluster x and C_x, and $d(c_i, c_j)$ is the c_i and c_j distance

$$\mathrm{BIC} = k\ln(n) - 2\ln(L) \tag{10}$$

where L is the likelihood of the given model data, k is the number of parameters in the model, and n is the sample size

$$\mathrm{CI} = 0.5\times\frac{\mathrm{DBI}-\mathrm{DBI}_{\min}}{\mathrm{DBI}_{\max}-\mathrm{DBI}_{\min}} + 0.5\times\frac{\mathrm{BIC}-\mathrm{BIC}_{\min}}{\mathrm{BIC}_{\max}-\mathrm{BIC}_{\min}} \tag{11}$$

where $\mathrm{DBI}_{\min}$ is the minimum value when the number of clusters is equal to three to eight, $\mathrm{DBI}_{\max}$ is the maximum value, $\mathrm{BIC}_{\min}$ is the minimum value when the number of clusters is equal to three to eight, and $\mathrm{BIC}_{\max}$ is the maximum value.

Finally, in the clustering results based on the optimal clustering parameters, the average SSAVI value for each class was calculated. Considering the characteristic high-value of SAV in the SSAVI image, the class with the highest mean value was identified as SAV [Fig. 2(c) and (d)].

C. Assessment of Algorithm

The accuracy and robustness of SSAVI-GMM algorithm are evaluated through two methods as follows.

1) Accuracy assessment based on validation datasets B and C. Confusion matrices were obtained between measured classes from validation sets and mapped classes derived from the SSAVI-GMM algorithm, and overall accuracy (OA), F1 score, user accuracy (UA), and producer accuracy (PA) were calculated [45].
2) Assessment by phenological feature of SAV. For lakes lacking SAV samples, indirect validation was conducted by phenological feature of SAV. Specifically, in a short timeframe, SAV exhibits fixed spatial distribution and limited area change. The similarity in spatial distribution of SAV between adjacent temporally sequential images was assessed by Dice coefficient [46] (12). In addition, the intraannual distribution of SAV also displayed seasonal variations

$$D = \frac{(2\times N_x)}{N}\times 100\% \tag{12}$$

where N_x is the count of pixels concurrently classified as SAV in two images, N is the combined count of pixels classified as SAV in two images, and D is the Dice coefficient.

IV. Results

A. Validations by Field Measured Data

In the five key lakes, the SAV maps were obtained by SSAVI-GMM algorithm based on Sentinel data, respectively (Fig. 4). The OAs were 85.71% in Lake Datong, 84.44% in Lake Shijiu, 87.72% in Lake Nanyi, 86.84% and 88.23% in Lake Taihu, and 88.89% in Lake Yangcheng [Figs. 4 and 5(a) and (b), Table I], respectively. Notably, the imagery on September 5, 2020 in Lake Taihu captured the outbreak of AB, with coexistence of AB [Region 1 in Fig. 4(c)] and FEAV [Region 2 in Fig. 4(c)]. In this complex environment, SAV can still be accurately extracted by the SSAVI-GMM algorithm, excluding FEAV and AB. In addition, the accuracy of classification maps from other 16 lakes in the Yangtze Plain was also evaluated by the validation dataset C, and OA was 88.28% (Table I).

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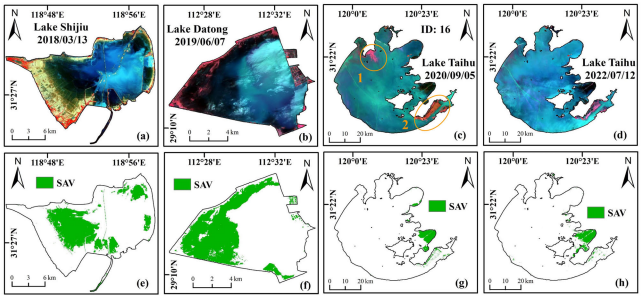


Fig. 4. SAV maps derived from SSAVI-GMM algorithm in Lake Shijiu, Lake Datong, and Lake Taihu. (a)–(d) False color images, and (e)–(h) corresponding SAV maps.

TABLE I
CONFUSION MATRIX OF VALIDATION DATASETS B AND C

Lake name	Class	SAV	Non-SAV	PA(%)	UA(%)	F1 score	OA(%)
Datong	SAV	36	1	85.71	97.30	0.91	85.71
	Non-SAV	6	6				
Shijiu	SAV	26	3	86.57	89.66	0.88	84.44
	Non-SAV	4	12				
Nanyi	SAV	42	2	85.71	95.45	0.90	87.72
	Non-SAV	7	8				
Taihu	SAV	34	4	89.47	89.47	0.89	86.84
	Non-SAV	4	19				
Taihu	SAV	91	3	89.22	96.81	0.92	88.23
	Non-SAV	11	14				
Yangcheng	SAV	41	1	89.13	97.62	0.93	88.89
	Non-SAV	5	7				
Validation dataset C	SAV	541	21	88.50	96.17	0.92	88.28
	Non-SAV	67	128				

B. Assessment by Phenological Changes

The SSAVI-GMM algorithm was applied to the imagery of Lake Nanyi on August 2 and August 5, 2018, as well as imagery of Lake Yangcheng on March 9 and March 14, 2022 (Fig. 5). The results indicated that the spatial distribution and area of SAV demonstrated a high level of overall consistency. Specifically, in Lake Nanyi, with a three-day interval between images, the SAV areas were 19.72 and 19.26 km^2 and the Dice coefficient was 79.01% [Fig. 5(a)]. In Lake Yangcheng, with a five-day interval between images, the SAV areas were 10.39 and 10.86 km^2 and the Dice coefficient was 92.38% [Fig. 5(b)]. There are still slight differences observed in SAV maps of different periods, and this may mainly be due to the growth and death of SAV itself.

Furthermore, the algorithm was used on images from different seasons in 2019 in Lake Datong [Fig. 5(c)]. The results demonstrated significant seasonal variations in SAV, attributed to diverse species and distinct vegetation patterns among different lakes [47]. Specifically, the SAV areas were 14.98 km^2 on April 8, 20.80 km^2 on August 16, 9.98 km^2 on September 30, and 14.14 km^2 on December 4 in Lake Datong, with the highest SAV areas in summer (August 16) [Fig. 5(c)].

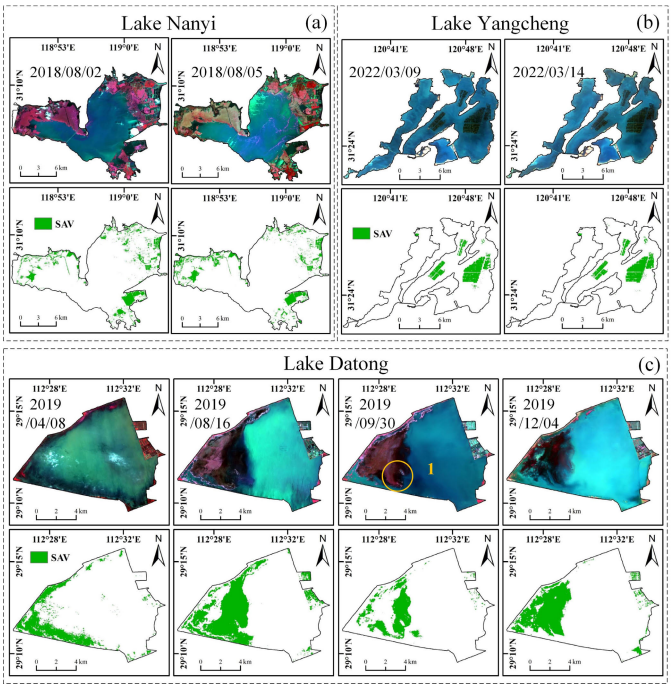


Fig. 5. (a) and (b) False color images with neighboring temporal and corresponding SAV maps in Lake Nanyi and Lake Yangcheng. (c) False color images in different seasons of 2019 and corresponding SAV maps in Lake Datong.

It is noteworthy that SAV in some regions [Region 1 in Fig. 5(c)] may be covered by floating aquatic vegetation on September 30, resulting in it not being detected by remote sensing.

C. Comparison With Submeter SAV Maps

We visually interpreted SAV distribution from the acquired submeter high-resolution JL-1 true color imagery on October 4, 2021 in Lake Taihu, September 22, 2021 in Lake Nanyi, June 7, 2021 in Lake Yangcheng, and October 22, 2020 in Lake Wuchang [Fig. 6(a1)–(d1) and (a2)–(d2)]. Then the SSAVI-GMM algorithm was applied to the imagery closest in time to the JL-1 image date for mapping SAV [Fig. 6(a4)–(d4)]. It is noteworthy that the SAV map in Lake Yangcheng was obtained from a closest date in the next year due to image quality issues. In comparison, the visual interpretation results are more detailed. In terms of the overall SAV distribution, both the results show a high level of consistency.

V. DISCUSSION

A. Comparison With the Existing Algorithm

The SSAVI-GMM algorithm developed in this study is the currently optimal algorithm for monitoring SAV on Sentinel-2. Although numerous decision tree algorithms for SAV mapping were developed, most of them were based on Landsat data, with fewer using Sentinel-2 data. The two most classic algorithms based on Landsat data are VBI algorithm [15] for mapping FEAV, SAV, and AB, and Dai et al.'s [14] algorithm for mapping SAV. Theoretically, they can also apply to Sentinel-2 data. However, when we transferred these two algorithms to Sentinel-2 MSI imagery to map SAV in

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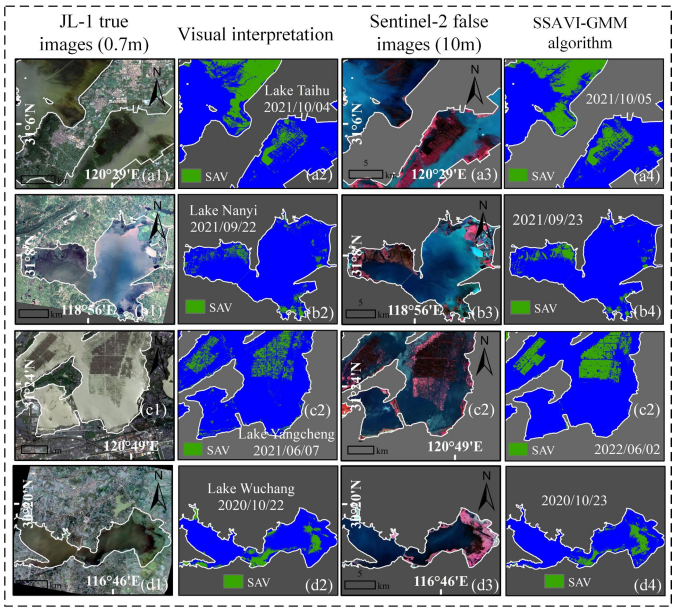


Fig. 6. Comparison of SAV maps derived from visual interpretation based on JL-1 images and SSAVI-GMM algorithm based on Sentinel data in Lake Taihu (a1)–(a4), Lake Nanyi (b1)–(b4), Lake Yangcheng (c1)–(c4), and Lake Wuchang (d1)–(d4).

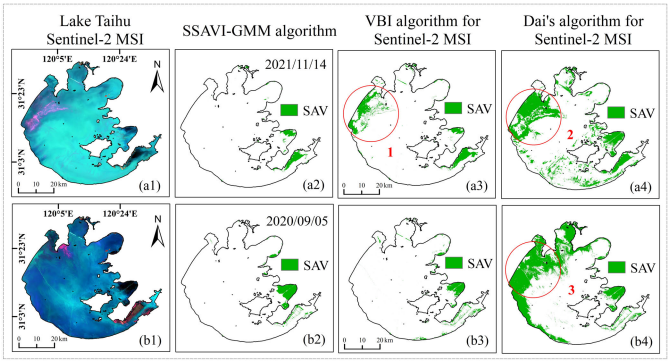


Fig. 7. Comparison of SAV maps derived from SSAVI-GMM, VBI, and Dai's algorithms based on Sentinel data in Lake Taihu. False color images (a1 and b1), and corresponding SAV maps derived from SSAVI-GMM (a2 and b2), VBI (a3 and b3), Dai's algorithms (a4 and b4). Note: VBI and Dai's algorithms were developed based on Landsat data. Here, we are applying them to Sentinel data for mapping SAV.

Lake Taihu, we observed that VBI and Dai's algorithms cannot always accurately extract SAV, especially in eutrophic lakes (Fig. 7). For example, three SAV classification maps were obtained by three algorithms based on Sentinel data on November 14, 2021 and September 5, 2020 in Lake Taihu, respectively. The results indicated that SAV was successfully identified using VBI algorithm based on the image of September 5, 2020 [Fig. 7(b3)], but AB was misclassified as SAV on November 14, 2021 [e.g., Region 1 in Fig. 7(a3)]. And AB was also misclassified as SAV by Dai's algorithm [e.g., Regions 2 and 3 in Fig. 7(a4) and (b4)]. In comparison, the SSAVI-GMM algorithm exhibited higher robustness, accurately mapping SAV in both Sentinel-2 imageries [Figs. 7(a2) and 8(b2)].

The SAV maps derived from the SSAVI-GMM algorithm based on Sentinel data are highly consistent with published results based on Landsat data [15]. For example, we obtained

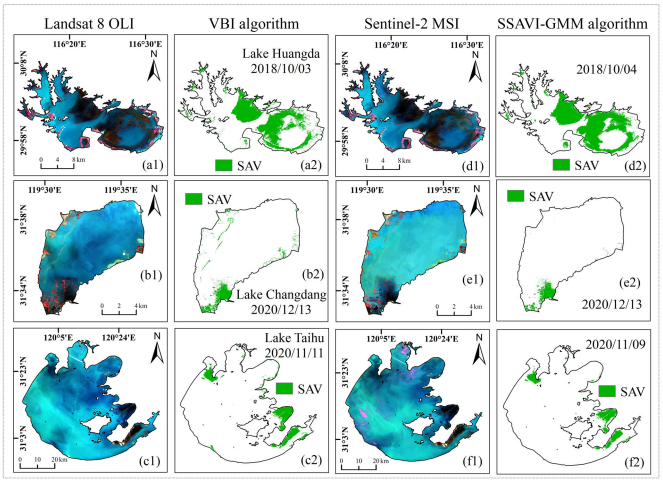


Fig. 8. Comparison of the reproduced SAV maps derived from VBI algorithm based on Landsat data and SSAVI-GMM algorithm based on Sentinel data in Lake Huangda (a1 and a2, d1 and d2), Lake Changdang (b1 and b2, e1 and e2), and Lake Taihu (c1 and c2, f1 and f2).

the SAV maps of Luo's study on October 3, 2018 in Lake Huangda, on December 13, 2020 in Lake Changdang, and November 11, 2020 in Lake Taihu [Fig. 8(a)–(c)]. Then the SAV maps closest in time to the published results date were obtained by the SSAVI-GMM algorithm, respectively [Fig. 8(d)–(f)]. Although the SSAVI-GMM algorithm and the published study use data sources with different spatial resolutions, the SAV distribution remains generally consistent. In addition, the SAV coverage derived from Luo's study and SSAVI-GMM algorithm in 2021 also remains consistent ($p < 0.001$, $R^2 = 0.8$) (Fig. S4).

B. Advantages and Limitations of Algorithm

The SSAVI-GMM algorithm proposed in this study is an innovative and excellent methodological sequence. On one hand, SSAVI constructed by fusing Sentinel-1 SAR and Sentinel-2 MSI data is significantly superior to other spectral indices. It can eliminate the interference of other lake cover types [Fig. 3(a)], enabling direct SAV mapping without the need to construct complex decision trees like traditional methods. And SSAVI also maintains certainly robust under varying water environments. For example, we identified the water with high chlorophyll-a concentration in the region of AB [e.g., Regions 2 and 3 in Fig. S5(b)] and the water with high suspended matter concentration at the estuary [e.g., Regions 2 and 3 in Fig. S5(b)] in the Sentinel-2 true color image on October 6, 2016. Although these two special waters were considered to have similar spectral features as SAV at certain wavelengths, especially chlorophyll-a [48], it is obvious they show significant differences from SAV in SSAVI and did not affect SAV mapping [Fig. S5(c) and (d)]. On the other hand, GMM clustering is critical to ensure the automation and robustness of SAV mapping in the SSAVI-GMM algorithm. In GMM clustering, the critical parameters and optimal models are determined by DBI and BIC and do not rely on field survey data or human interventions. Notably, unlike traditional threshold-based decision tree methods, which are sensitive

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to threshold variations in classification results, the GMM approach shows minimal fluctuations in SAV area once a specific cluster count is achieved, with fluctuations below 0.1% in amplitude (Fig. S6). Thus, classification results based on GMM clustering are notably more robust and accurate.

The SSAVI-GMM algorithm still two limitations. First, due to disparate temporal resolutions of Sentinel-1 SAR and Sentinel-2 MSI data, the time interval between the images used for fusion directly impacts the SSAVI. Therefore, careful consideration must be given to the influence of the time interval between optical and SAR data used for fusion. We quantitatively evaluated the impact of the time interval of fused images on SSAVI quality using OA, information entropy, and peak signal-to-noise ratio [49] (Fig. S7). The results indicate that selecting SAR images for fusion with an optical image at the closest time interval (around ten days) is optimal, ensuring an error margin in 10%. In addition, the SSAVI-GMM algorithm also struggles to address the black water, like all other current SAV mapping algorithms (e.g., Regions 1 and 5 in Fig. S5) [15]. The extreme anoxic condition of black water results in the accumulation of significant amounts of dissolved organic matter. The high concentration of chromophoric dissolved organic matter (CDOM) leads to low spectral reflectance, particularly in the visible bands [50]. The similarity of black water in spectral and imaging characteristics with SAV is the main reason for the misclassification in the SSAVI-GMM algorithm, as well as in other algorithms.

C. Potential Implications of Algorithm

Global SAV significantly declined due to factors such as variations in underwater light conditions and nutrient enrichment [8], [51]. Simultaneously, studies indicated that more than 60% of lakes worldwide were considered eutrophic [52], with 68% experiencing ABs, and over 8.8% undergoing algae proliferation [53]. Does this global phenomenon of SAV degradation and ABs signify a transition from clear-water state dominated by SAV to turbid water state dominated by algae in lakes? To answer this question, extensive and high spatiotemporal resolution data of SAV are imperative. The SSAVI-GMM algorithm proposed in this study has already demonstrated its high accuracy and robustness for mapping SAV in various complex lake environments. Based on the SAV data offered by SSAVI-GMM algorithm, in the future, we can explore the stable state transitions and assess the health status in the global shallow lakes.

In addition, as a primary producer commonly present in inland shallow lake ecosystems, SAV has the capacity to absorb and sequester a significant amount of CO_2 [54], [55]. Nevertheless, due to the lack of SAV data in current assessments of lake carbon sources and sinks, SAV areas have been largely neglected over the long term [56], [57]. For instance, in a previous study, CO_2 fluxes were only estimated by MODIS data in the zone dominated by cyanobacteria of Lake Taihu [56]. This estimation, which did not take into account the zone dominated by SAV in Lake Taihu, evidently introduced uncertainty into the overall study of lake carbon flux. Therefore, the introduction of SSAVI-GMM algorithm provides significant support for conducting carbon flux estimation study in SAV area, which has been challenging due to the lack of SAV data.

VI. Conclusion

In this study, we developed the SSAVI-GMM algorithm in a typical phytoplankton–macrophyte coexistence lake (e.g., Lake Taihu). The key index of algorithm is SSAVI, which is exclusively sensitive to SAV, and constructed by fusion of Sentinel-1 SAR and Sentinel-2 MSI images. SSAVI was then combined with GMM clustering to achieve automatic map SAV. We tested SSAVI-GMM algorithm in 19 lakes of the Yangtze Plain and the average OA is 87.15%. In addition, we also assessed SSAVI-GMM algorithm by SAV phenological feature and submeter high-resolution imagery, as well as compared with two classical algorithms. The results showed that our algorithm demonstrated excellent robustness and accuracy. We believe that the SSAVI-GMM algorithm has the potential to provide high spatiotemporal SAV data in shallow lakes globally.

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